Soc Lab

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SoC Architecture:

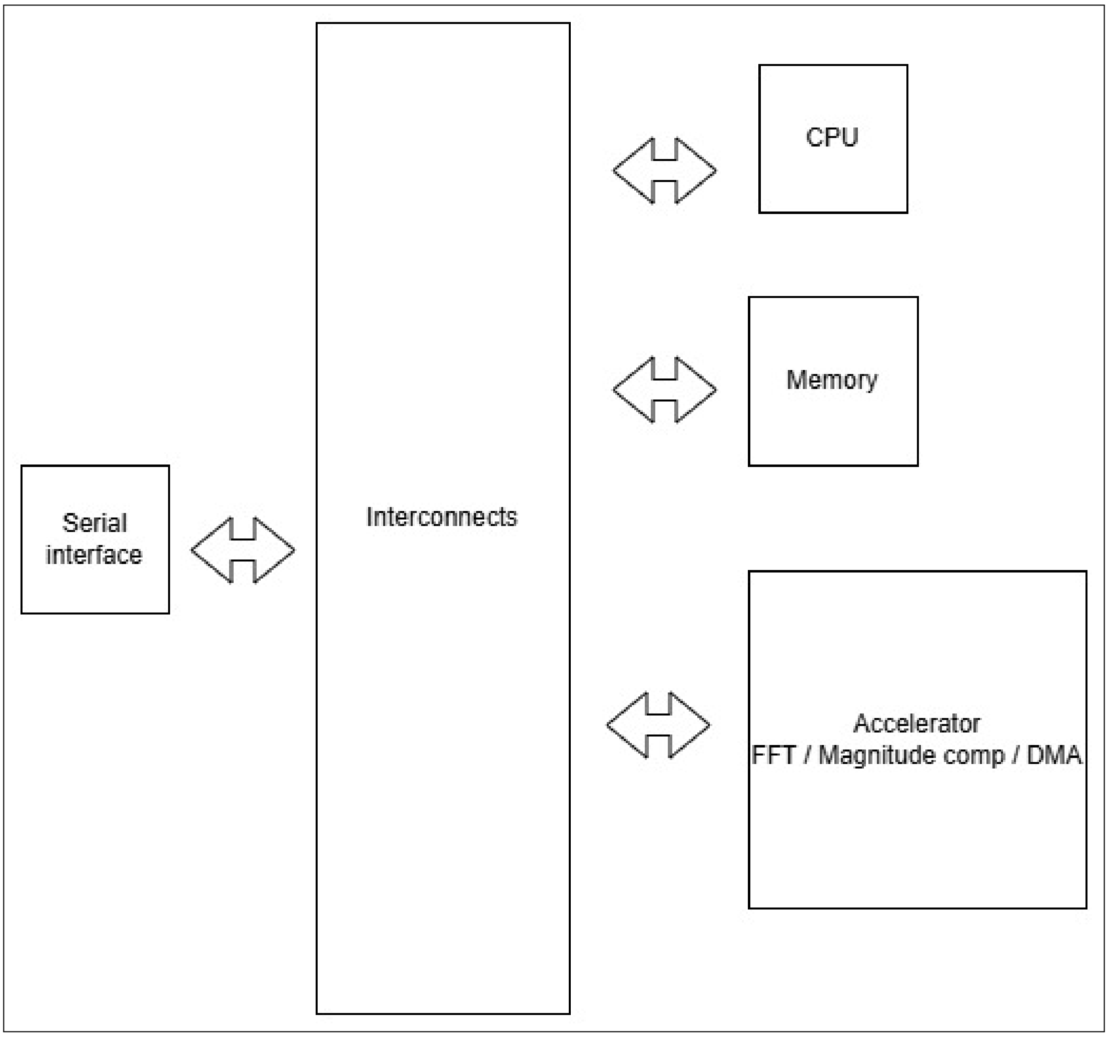

Figure 1: SoC Architecture

SW HW Partitioning:

- **Store samples in memory:** HW using the DMA to avoid CPU overhead whenever an interrupt from the UART arrives.
- **Apply optional windowing:** SW as it is optional and configurable.
- **Compute FFT:** HW using the FFT accelerator as it requires complex computation and it is repetitive.
- **Compute magnitude spectrum:** HW using magnitude accelerator as it requires multiplication, addition and squaring. It is also repetitive.
- **Find peak bin:** SW as it requires simple sorting algorithm.
- **Convert peak bin to frequency:** SW as it is not computationally intensive. It is also done once.

Pseudocode:

1. Interrupt arrives.
2. Signal DMA to store data in memory.
3. Apply windowing.
4. Signal FFT accelerator to start working.
5. Receive a signal from the FFT accelerator that it finished its job.
6. Store output of the FFT accelerator in memory.
7. Signal magnitude accelerator to start working.
8. Magnitude accelerator fetches data from memory and computes magnitude.
9. Magnitude accelerator stores its output in memory and signals the CPU that it finished.
10. CPU fetches data from memory and applies sorting algorithm to find the peak.
11. CPU converts the peak bin to frequency.
12. It sends output to the serial interface.